Grocery Price Prediction and Analysis

Milestone: Final Submission

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PROBLEM SETTING

Inflation and the rising costs of food have become one of the most important concerns in the United States, driving people to political discourse. Americans have become increasingly unable to pay for their higher costs of living, forcing them to choose which products to buy from grocery stores, and which ones to leave on the shelves. Certain households show sentiments of skipping meals in order to stay within the means of their grocery budgets. This financial strain is not only the result of groceries, but from a rising cost of living driven by housing, utilities, transportation, and other goods and services. The global COVID-19 pandemic served as a catalyst of these issues, causing supply-chain disruptions, increased inflation, labor shortages, and demand fluctuations. Lower-income households oftentimes face the most financial strain, leading to racial biases when needing assistance. Understanding the patterns of goods can be beneficial for the consumer to make more informed decisions about their grocery purchases, and for lawmakers to address food costs.

PROBLEM DEFINITION

This analysis seeks to understand the fluctuations in the Consumer Price Indexes of some of the most popular grocery store prices. Specifically, we plan to model how grocery prices had trended before the COVID-19 Pandemic and forecast price behaviors excluding pandemic influence. The affordability of groceries will also be evaluated. Key questions that we plan to investigate include:

- Prior to the pandemic, what were the trends in grocery prices?
- Based on pre-pandemic trends, how would forecasted prices look like?
- Using the most updated prices, which items have deviated the most from pre-pandemic prices?
- What grocery items tend to have similar price behaviors with others?
- How affordable are groceries on a household's income, and how does race play a factor?
- What other observations could be made?

DATA SOURCES

We will be using ten datasets that show the average monthly price of some of the most bought products from 2015-2024. A separate dataset will have the monthly Consumer Price Indexes of all prices found at grocery stores (Food at Home). These datasets are all U.S. City Averages, not adjusted for seasonality. These datasets are downloaded from the U.S. Bureau of Labor Statistics.

Reference:
Bureau of Labor Statistics, U.S. Department of Labor, Consumer Price Index Average Price Data,
http://data.bls.gov/dataViewer/view/timeseries/APU0000704111,
https://data.bls.gov/dataViewer/view/timeseries/APU0000702111,
https://data.bls.gov/dataViewer/view/timeseries/APU0000FF1101,
https://data.bls.gov/dataViewer/view/timeseries/APU0000717311,
https://data.bls.gov/dataViewer/view/timeseries/APU0000708111,
https://data.bls.gov/dataViewer/view/timeseries/APU0000703112,
https://data.bls.gov/dataViewer/view/timeseries/APU0000709112,
https://data.bls.gov/dataViewer/view/timeseries/APU0000712112,
https://data.bls.gov/dataViewer/view/timeseries/APU0000701312,
https://data.bls.gov/dataViewer/view/timeseries/APU0000703613,
http://data.bls.gov/dataViewer/view/timeseries/CUUR0000SAF11,
accessed on January 23, 2025.

We will also use the annual household incomes in the U.S. from the United States Census Bureau.

Reference:
U.S. Census Bureau, "Table A-2. Households by Total Money Income, Race, and Hispanic Origin of Householder: 1967 to 2023", Income in the United States: 2023, https://www.census.gov/data/tables/2024/demo/income-poverty/p60-282.html, accessed on January 24, 2025.

DATA DESCRIPTION
Each dataset from the Bureau of Labor Statistics (BLS) has the following format:

Column	**Description**
Series ID	ID of Time Series
Year	Year (2015-2024)
Period	Month (M01-M12)
Label	Year/Month Label (ex: 2017 Jan)
Value	Consumer Price Index (CPI) value for related year/month

The dataset from the U.S. Census Bureau has the following format:

Column	Description
Race and Hispanic origin of householder and year	Separates data by year, grouped by race
Number (thousands)	Number of people surveyed
Percent distribution	Divided into 9 sub-columns to represent different income brackets (Under $15,000, $15,000 to $24,999, $25,000 to $34,999, $35,000 to $49,999, $50,000 to $74,999, $75,000 to $99,999, $100,000 to $149,999, $150,000 to $199,999, and $200,000 and over)
Median income (dollars)	Divided into 2 sub-columns to represent median income and margin of error
Mean income (dollars)	Divided into 2 sub-columns to represent mean income and margin of error

METHODOLOGY

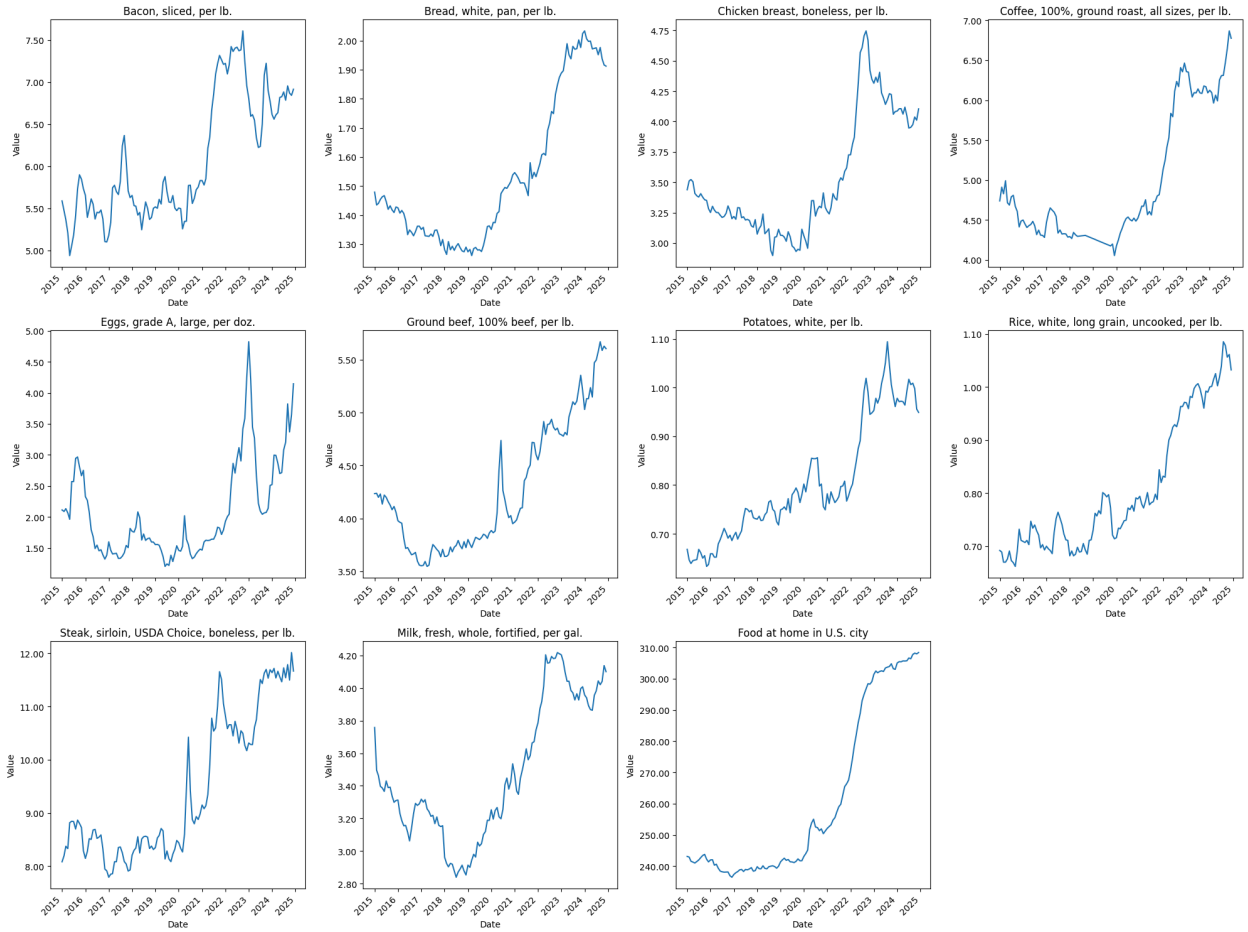
For this project, an ARIMA time-series model will be used to forecast grocery prices from 2020 to 2024 based on the years 2015 to 2019. We will be using the monthly consumer price indices of ten grocery items and the 'Food at Home' consumer index to forecast price trends without regarding the pandemic disruption. Additionally, we will conduct price correlation analysis between items using several techniques including correlation matrix, dendrogram, and Principal Component Analysis (PCA). We will also use the affordability index to analyze the burden of grocery prices on a household's income, as well as the relative affordability index to determine if any disparities exist between a racial group and the financial burden of grocery prices.

BLS Dataset

Data Preprocessing & Preliminary Visualization

Our BLS series were imported to our Jupyter file using the Python glob import and concatenated into a single DataFrame. This DataFrame consisted of Series ID, Year, Period, Label, and Value. The DataFrame was then standardized for a consistent format

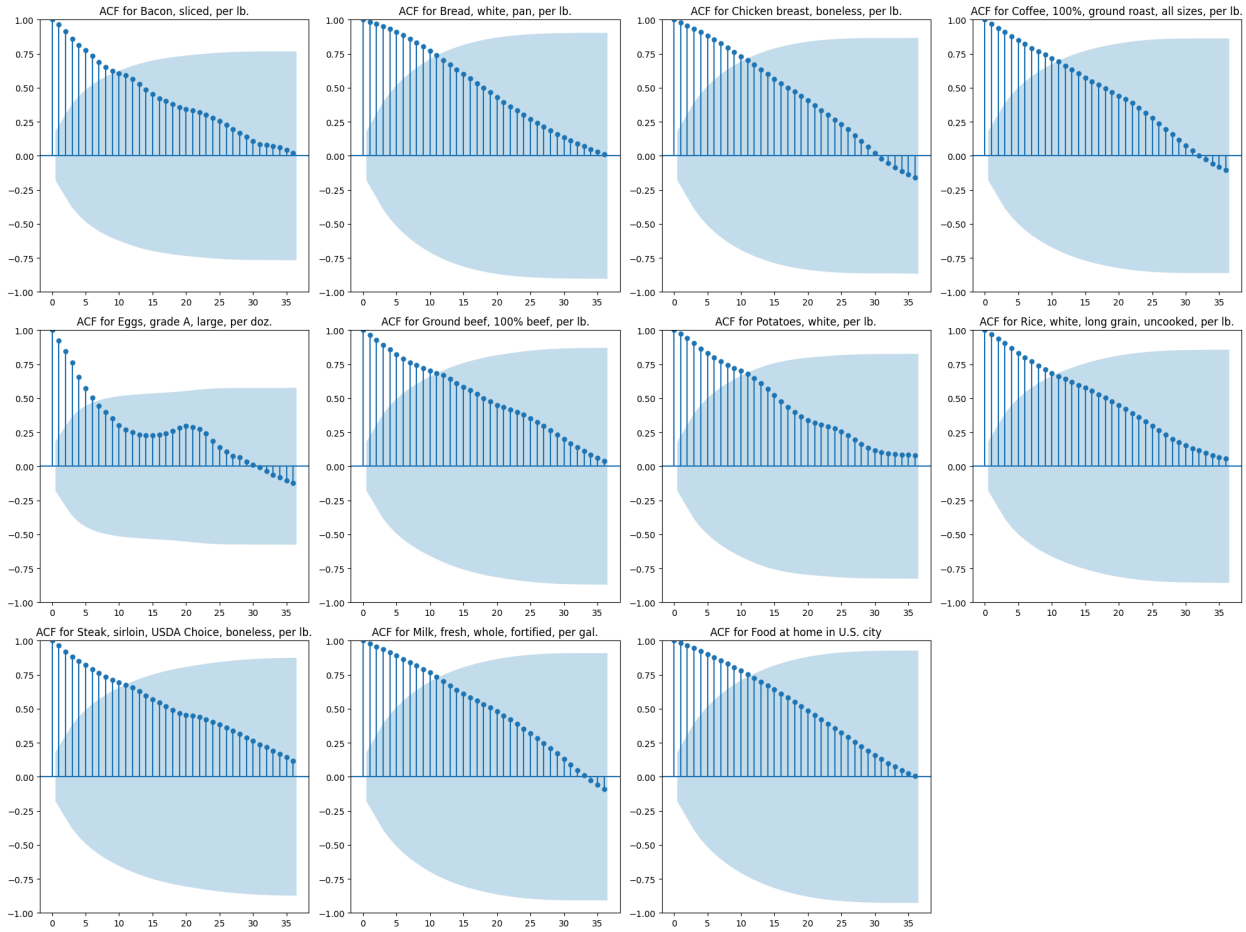
that could be used through all steps of the project. Labels were changed to a standardized datetime format, series IDs were mapped to manageable names, and repetitive columns were dropped. The data was then pivoted using the datetime index, and missing values in two series were imputed using linear interpolation. The plots below showcase the values of each grocery item by date.



By having a preliminary look on each series, we notice that there is a trend in each product. For all products, a drastic increase in prices occurred at some point in or after 2020. This could be attributed to supply and demand created by the COVID Pandemic, although more analysis is needed to determine a cause-effect relationship.

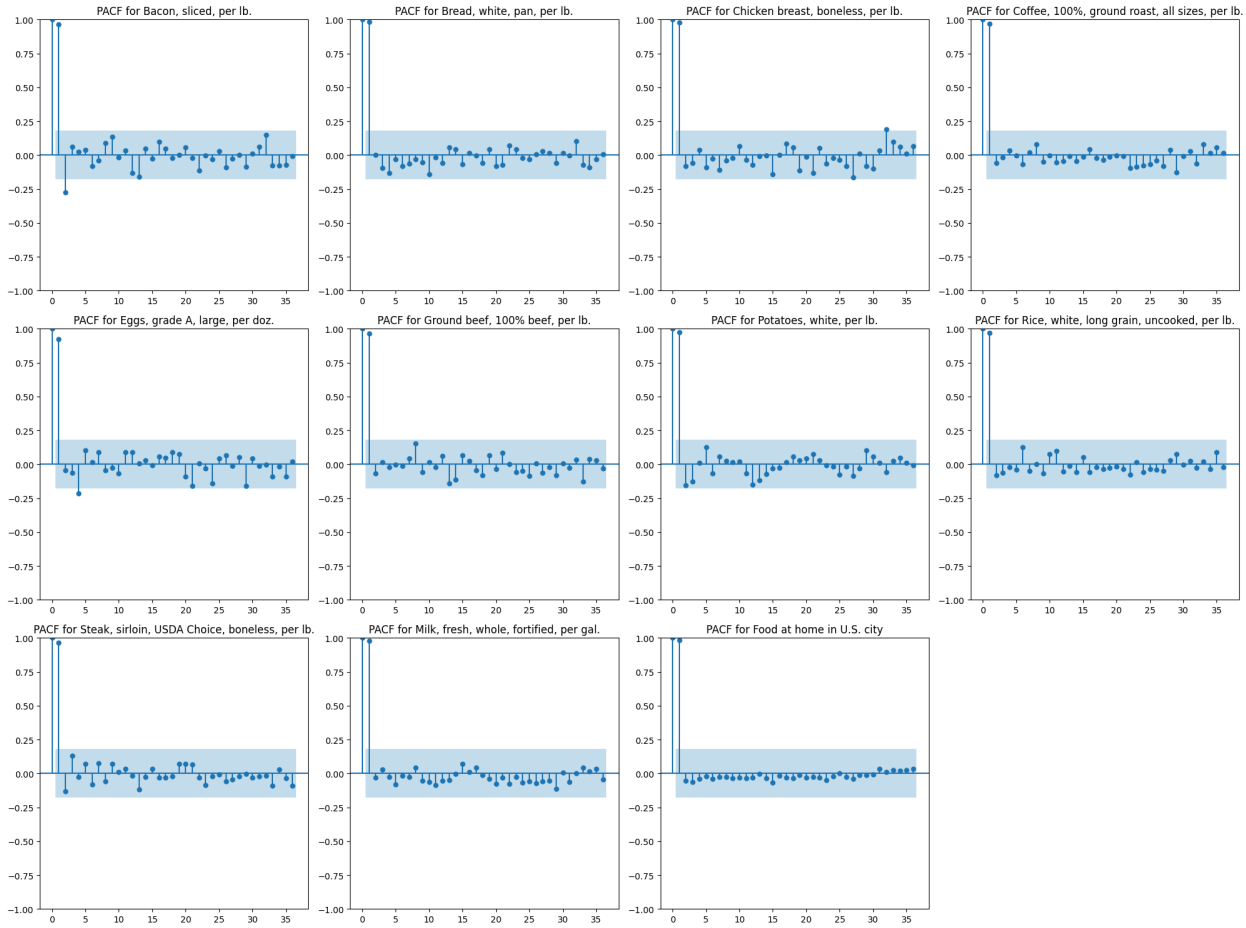
Model Implementation

Our primary model used is an ARIMA time-series model. First, an autocorrelation function (ACF) plot for each item was created to determine seasonality in each series.



There are no clear seasonal spikes at fixed intervals, suggesting no seasonality. The prices of food items show a gradual decay in correlation over time.

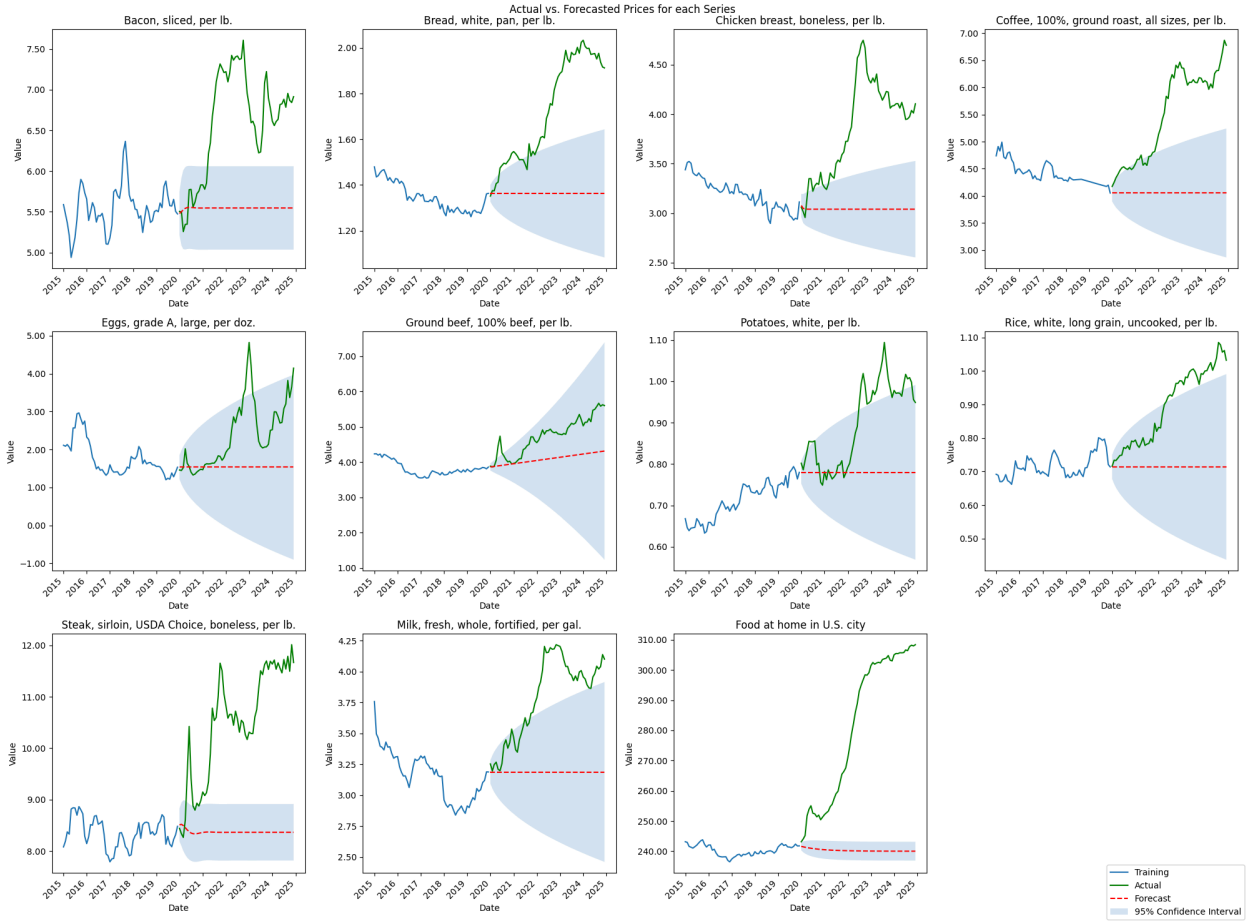
A partial autocorrelation function (PACF) plot was then created.



The PACF plots have a sharp cutoff after the second lag. This suggests that the immediate past value strongly influences the next value, and a low-order autoregressive model may be suitable for time-series modeling.

With these two plots as well as the clear trends from the preliminary data visualization, we can determine that the data appears close to stationarity within the confidence interval, and with no clear seasonal spikes.

The ARIMA model was then fitted and prices were forecasted for each food item series. For simplicity, our model forecasts prices from the first month of 2020 as a marker for the COVID-19 pandemic. Data from 2015-01-01 until 2019-12-01 were used as a training set, and data from 2020-01-01 onwards were used as the test set. The auto_arima function from the pmdarima library was used to automate the (p, d, q) order selection. A 95% forecast confidence interval was included in the plotting of the model. The ARIMA models for each series are shown below.



Certain food items, such as Ground Beef, Potatoes, and Rice, presented with low-to-moderate Mean Absolute Errors (MAE), Mean Squared Errors (MSE), and Root Mean Squared Errors (RSME). Along with these error metrics, the actual prices of these items from 2020-2024 mostly stayed in, or around, the 95% confidence interval of predicted prices. This means that the ARIMA model was fitted decently well within these items. This could be attributed to more stable supply chains or predictability in consumer demand, leading to price stability.

On the other hand, items such as Sirloin Steak, Eggs, and Coffee were fitted more poorly to the ARIMA model as they have high forecast errors, along with prices that deviate greatly from the confidence intervals our model created. Our consumer price index values for all groceries in our model are highly inaccurate compared to the actual values. These large errors may be attributed to issues that arose from the pandemic, including supply and demand, labor and production disruptions, supply chain issues, etc. 'Food at Home' specifically represents the CPI of groceries, and indicates that a large shift in grocery prices occurred beginning in 2020.

Although the ARIMA model gave us a good indication of COVID-19 influences, a more in-depth analysis into these exogenous variables is needed to determine a causal effect of the pandemic on grocery prices.

Model Evaluation

Each ARIMA model was evaluated using the Mean Absolute Error (MAE), Mean Standard Error (MSE), and Root Mean Squared Error (RMSE), as well as in percentage formats.

In general, an MAE of 4.49 (20.01%) suggests a moderate error across all items. An MSE of 206.56 (96.14%) is a very high error that suggests models have large outliers that are affecting its performance. The RMSE value 5.20 (23.92%) suggests that the discrepancies in models could be inflated by larger individual errors.

Individual breakdowns are as the following:

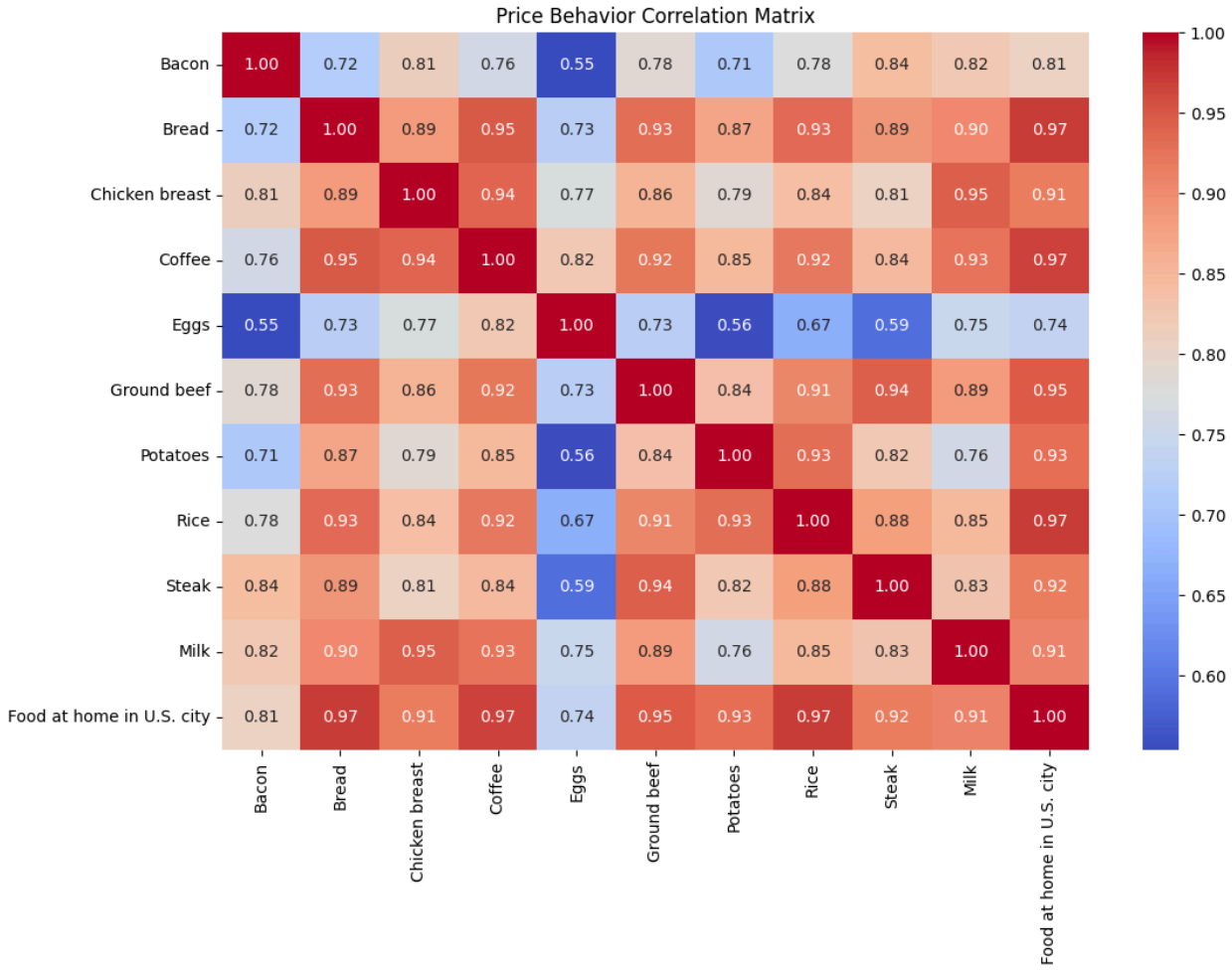
- The errors from the Bacon model are in the lower-moderate range, indicating that the forecasted prices are slightly reliable.
- The RMSE value of 0.42 (24.53%) in the Bread model suggests that some outliers are affecting the performance of the model, though the forecast has a relatively good fit.
- There are moderate error percentages for the Chicken Breast model, meaning that the prices of chicken could be improved, but are still somewhat reliable depending on the context.
- All error metrics for Coffee lead to large inaccuracies in forecasting prices, meaning the model struggled in accurately predicting Coffee prices.
- The error metrics for Eggs are larger than the previous series (Coffee), also suggesting inaccuracies in prices. This could be attributed to other external factors such as the Avian Flu.
- The low-to-moderate range of errors in the Ground Beef forecasted prices suggests a good fit to the model.
- The MSE value of 0.02 (2.62%) in the Potatoes model suggests very small errors, with a moderate discrepancy from individual errors. This model is a good fit.
- Similar to the chicken breast model, the Rice predictions are somewhat reliable in accuracy. The MSE value of 0.05 (5.08%) indicates that the predicted values are really close to the actual values, though there is a moderate discrepancy due to individual errors.
- Poor performance is indicated in the Steak model due to large prediction errors that inflate the MSE.

- The forecasts on Milk prices are relatively accurate.
- The MSE 2258.59 (802.60%) for Food at Home is past the point of consideration. This category cannot forecast food prices accurately due to this large error. Other factors may have caused this model to fit poorly.

Price Correlation between Food Items

Three techniques were used to determine correlations between food items, including a correlation matrix, dendrogram, and a principal component analysis (PCA) chart.

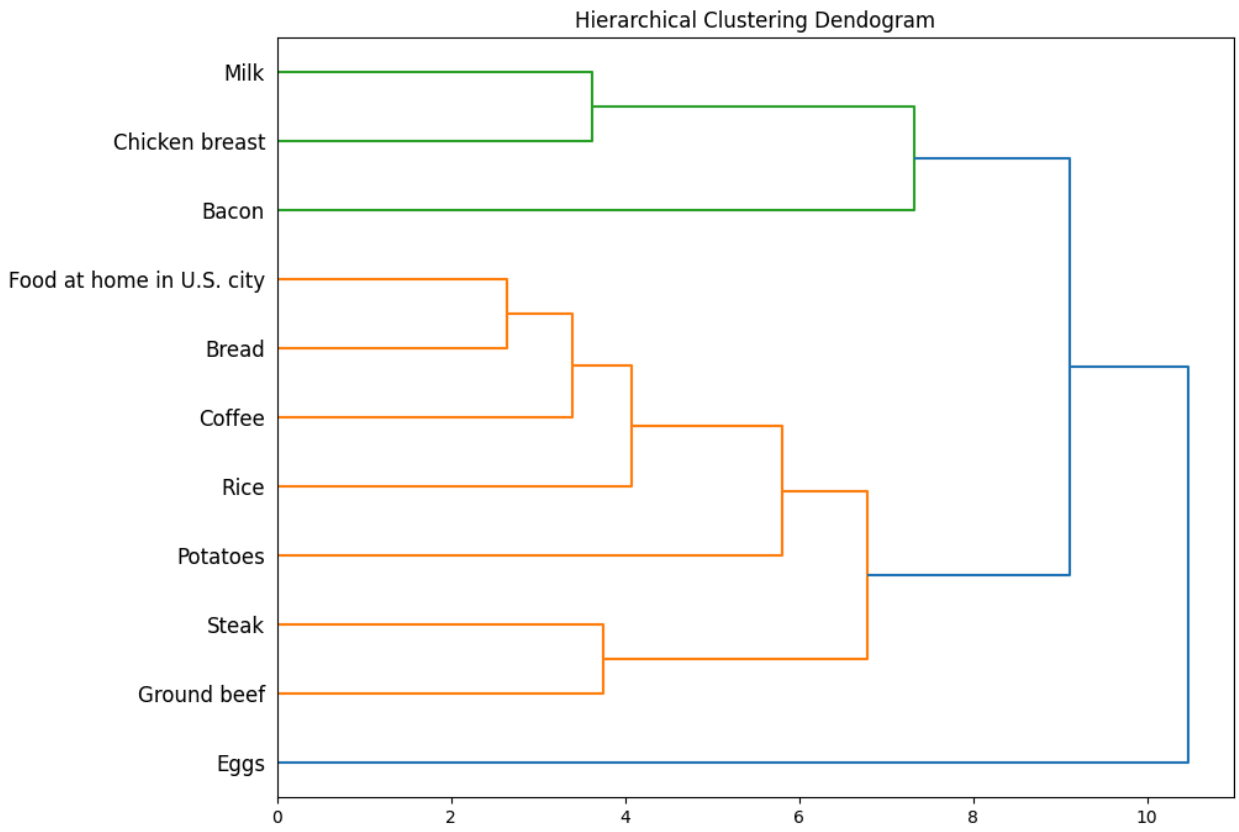
The correlation matrix is the following:



Foods that are highly correlated with the 'Food at Home' series suggest that these items are staples in American households and are subject to price fluctuations based on market trends. Other strong correlations may be influenced by consumer demand shifts based on the economy. Specific correlations such as Ground Beef & Steak, Bread & Rice, and Rice & Potatoes could be attributed to the food groups they are a part of, subject to exogenous variables such as labor costs and weather conditions for agricultural trends.

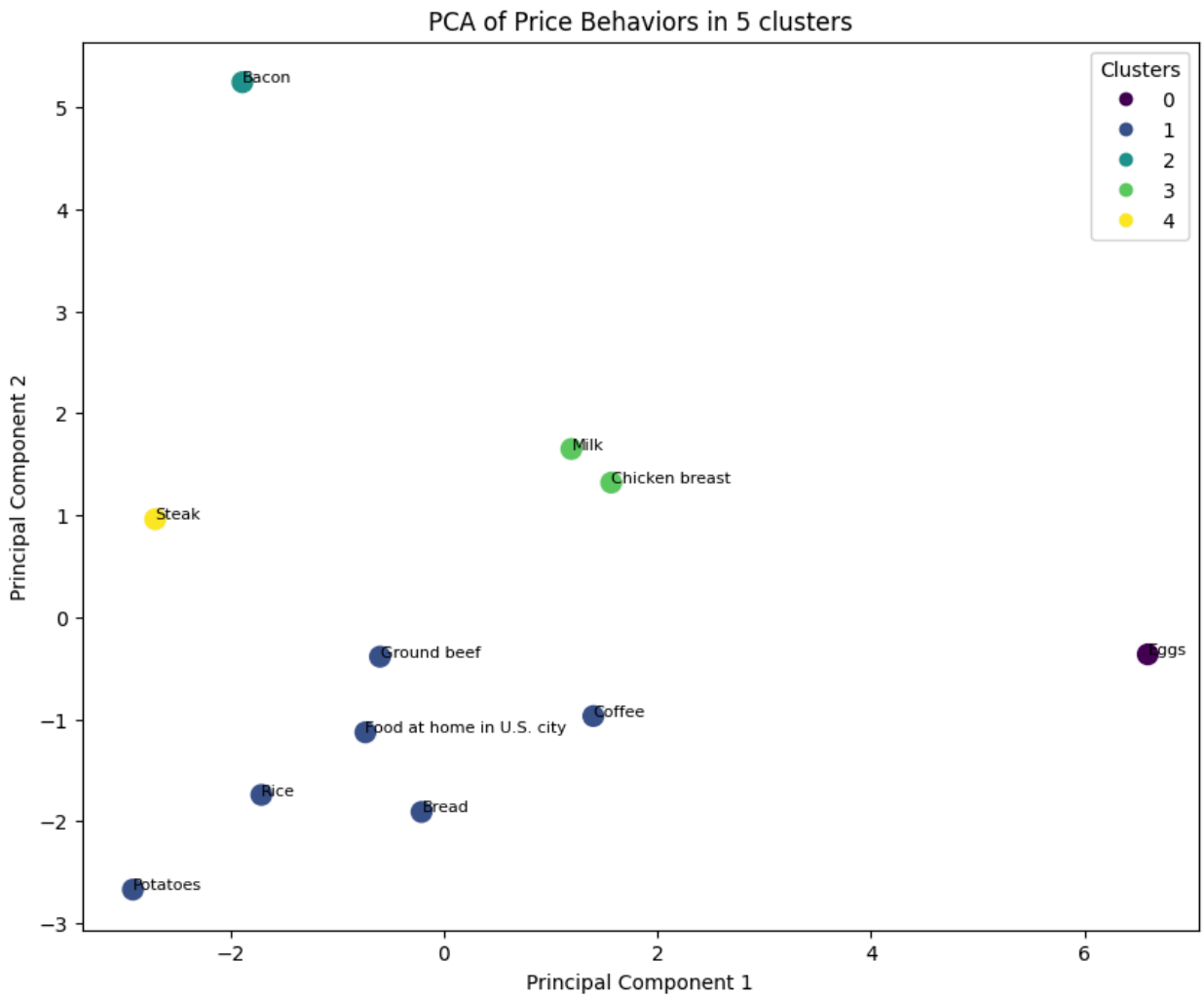
Eggs are the basis of the weakly correlated price behaviors, and could be attributed to the avian flu and feed costs due to rising grain prices. Eggs come from entirely different sources than their pairs, and have a completely separate supply chain and forms of consumption.

The dendrogram was then created for hierarchical clustering.



The dendrogram seems to suggest five major clusters that best fit our data. From the dendrogram, we notice that there is a strong similarity in prices between Chicken Breast and Milk, linking poultry and dairy together. Similarly, Ground Beef and Steak are clustered as beef prices. Food at Home, Bread, Coffee, Rice, and Potatoes are all linked to similar price behaviors. The price patterns of Bacon are separated, and the prices of Eggs are completely independent from the other prices.

A PCA plot was created using an interactive slider to adjust for the number of clusters. Similarly to the dendrogram, our food items are best clustered with five clusters.



Using Principal Component Analysis (PCA), our data suggests Eggs, Steak, and Bacon have isolated price behaviors. Milk and Chicken Breast are grouped as poultry and dairy. Potatoes, Rice, Bread, Coffee, Ground Beef, and our Food at Home index are all grouped together in one cluster, suggesting that these are staple items.

Census Dataset

Census Data Parsing

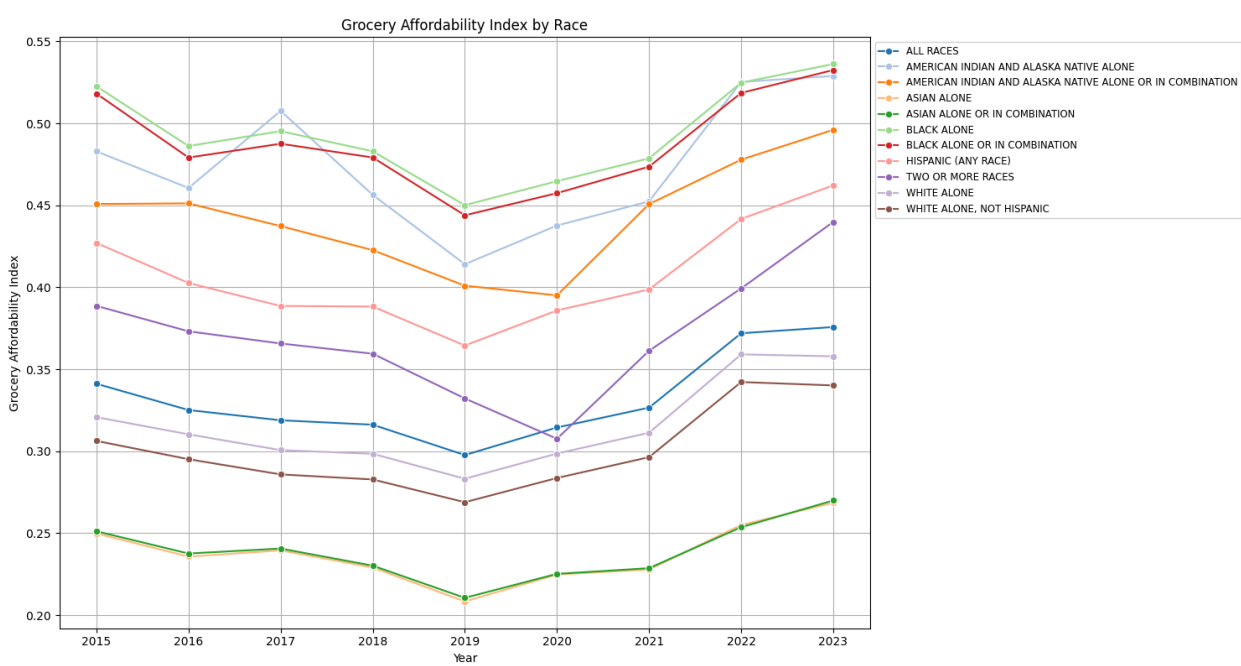
The .xlsx dataset provided by the U.S. Census contains information on households by income and race from the years 1967 to 2023. For the purpose of this project, only the years 2015-2023 are being used to compare grocery affordability to household income. It is assumed that the estimated median income (in dollars) is representative of the average household income of that year. The estimated household income between the

years provided are extracted for each racial group. After pivoting, the DataFrame is seen as the following:

Race Year	ALL RACES	AMERICAN INDIAN AND ALASKA NATIVE ALONE	AMERICAN INDIAN AND ALASKA NATIVE ALONE OR IN COMBINATION	ASIAN ALONE	ASIAN ALONE OR IN COMBINATION	BLACK ALONE	BLACK ALONE OR IN COMBINATION	HISPANIC (ANY RACE)	TWO OR MORE RACES	WHITE ALONE	WHITE ALONE, NOT HISPANIC
2023	80610	57270	61060	112800	112200	56490	56880	65540	68860	84630	89050
2022	77540	54910	60360	113100	113700	54960	55620	65300	72230	80320	84280
2021	79260	57220	57430	113600	113200	54080	54660	64930	71650	83160	87340
2020	79560	57170	63330	111300	111100	53840	54700	64840	81340	83800	88200
2019	81210	58380	60290	116000	114800	53710	54460	66330	72770	85350	89900
2018	75790	52510	56700	104600	104100	49620	50010	61720	66660	80300	84740
2017	74810	47000	54550	99600	99130	48170	48930	61390	65230	79340	83450
2016	73520	51900	52980	101400	100600	49170	49890	59370	64060	77030	80990
2015	71000	50160	53730	96940	96430	46350	46750	56720	62320	75510	79080

Affordability Index Calculation & Visualization

The affordability index is a metric used to measure the affordability of certain goods and services. In the context of groceries, we use the 'Food at Home' consumer price index as the generalized household grocery burden. To connect our racial data with grocery prices, the consumer price index is converted from monthly indices to an annual average. The affordability index is then defined as the radio of the average grocery index to household income by race, multiplied by 100. Each index is plotted by race afterwards.



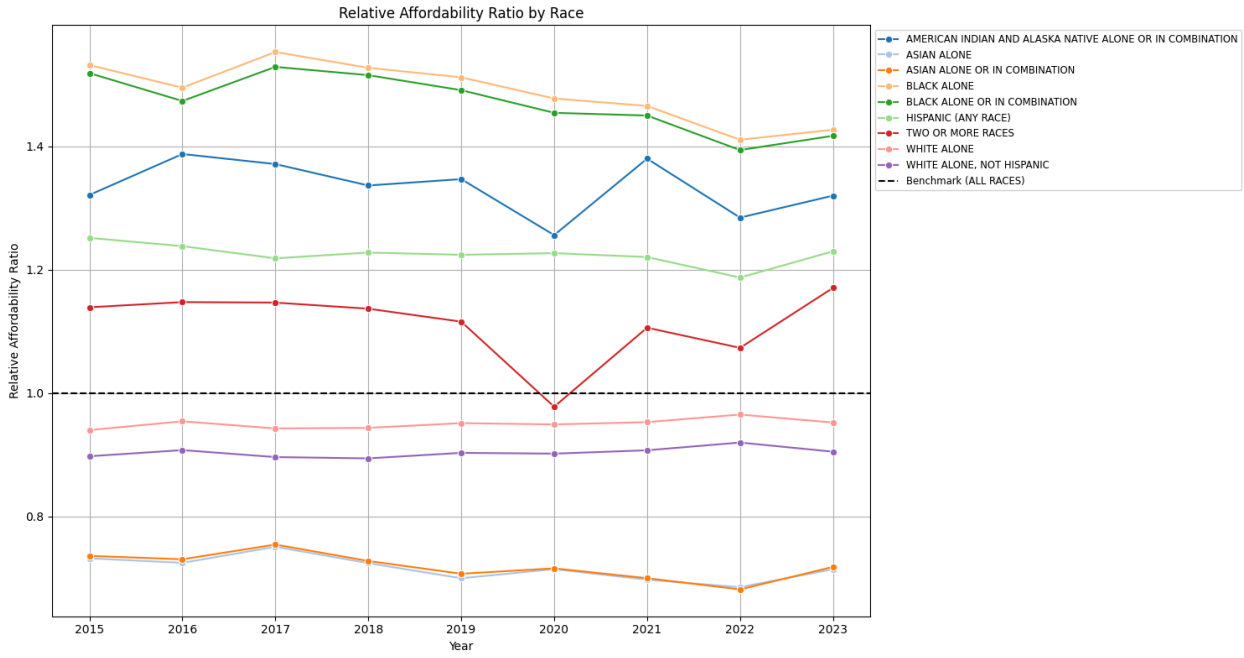
Affordability Index Interpretation

By analyzing the affordability index metric against grocery prices from 2015-2023, we notice a relatively decreasing trend from 2015-2019, followed by an upwards trend from 2019-2023. This indicates that grocery prices were becoming more affordable to the U.S. household until 2019, followed by worsening affordability onwards. This suggests that the COVID-19 Pandemic had economic impacts on all households.

For all racial groups, groceries are becoming a larger portion of household expenses. Black and American Indian and Alaskan Native populations consistently experienced the most financial burden when purchasing groceries. Both these groups had an approximate 2x more burden than the Asian populations. Asians and White populations experience the least burden when buying groceries.

Relative Affordability Calculation & Visualization

The relative affordability of groceries was calculated by racial group in comparison to the average U.S. household, designated as the race ‘ALL RACES’. This is calculated as the ratio of the affordability index of each racial group by the benchmark race. When plotted, racial groups are plotted on a scale relative to the ‘ALL RACES’ group, which has a ratio of 1.0.



Relative Affordability Interpretation

The benchmark dashed line at 1.0 represents the burden of the average U.S. household. The Black Alone and/or in Combination groups are between 1.4 and 1.6, meaning that these groups face 40-60% more financial burden with groceries than the average household. Hispanic and American Indian and Alaskan Native households face between 20-40% more burden. Asian and White populations consistently face lesser burdens than the average household, which showcases less pressure on their incomes.

CONCLUSION

Summary of Findings

Analyzing grocery prices not only confirmed the disruptions in grocery prices caused by the COVID-19 Pandemic, but also displayed broader issues in terms of income inequality. Between 2015 and 2019, the prices of certain groceries were more predictable, as the prices of the previous month were likely to influence the following month. However, ARIMA model forecasts from 2020 onwards deviated from actual prices, highlighting unprecedented volatility caused by the pandemic. The prices of certain grocery items post-pandemic, such as Eggs, Sirloin Steak, and Coffee, were at least 20% higher than projected. Other items such as Rice, Potatoes, and Bread remained relatively price stable. Clustering analysis identified five major pricing groups dependent on supply chain sensitivity. Conducting affordability analysis revealed higher income burdens on Black and Native American and Alaskan Native households, nearly double the burden than in Asian households. These disparities stayed consistent throughout the years, detailing not only the increasing burden of groceries on a household's income due to the COVID-19 Pandemic, but also the racial disparities suggesting systemic inequality. Monitoring the affordability index could guide adjustments to assistance programs and inflation relief.

Limitations

Although this analysis offers a preliminary insight into how grocery prices were affected by COVID-19, there are several limitations to be acknowledged. First, a direct causal-effect relationship between the pandemic and grocery prices cannot be determined without incorporating other external factors such as supply chain disruptions and inflation metrics into the model. Using the ARIMA model as-is for this reason would rely on the assumption that past events could predict future values, which is not the case. Adding additional factors to a SARIMAX forecasting model would provide clearer results. Second, consumer index metrics and estimated incomes are only an aggregation and may not accurately reflect household patterns such as household size, employment statuses, or spending habits. Federal and state assistance programs are

also not accounted for, nor is the variability of cultures within racial groups. Despite these limitations, the analysis provides a general direction on understanding the impacts of the COVID-19 Pandemic on grocery inflation.